
DEALER POSITIONING & HEDGING FLOWS (GEX)

Jonas Löffler

loefflerjonas6@gmail.com

v1.0 · September 20, 2026

Abstract

Gamma exposure (GEX) has become the most popular lens on options-driven price dynamics and the least disciplined. This paper builds the dealer-hedging story from the ground up and then states plainly how much of it freely available data can and cannot support. We begin with the market maker's business and the inventory problem that forces mechanical delta-hedging; we then confront the *identification problem* before writing a single formula. Open interest, volume, and implied volatility are observable; who is long and who is short is not, and the standard positioning assumptions that convert one into the other break in the meme-stock and 0DTE regimes. We construct GEX on a real, free CBOE chain snapshot, show that three defensible estimators place the gamma flip within a third of a percent of spot yet disagree on its *sign*, and derive the charm and vanna re-hedging flows analytically. Because free historical strike-level open interest does not exist in this pipeline, we run an explicitly labelled proxy regime study and a pre-committed pinning test rather than fabricate a GEX time series, and we translate each mechanical claim into a pre-registered, falsifiable test. Every number uses free data and is reproducible.

Keywords — Gamma exposure, GEX, dealer hedging, market maker inventory, delta-hedging, gamma flip, zero gamma, charm, vanna, 0DTE options, pinning, options open interest, identification problem, volatility clustering, leverage effect, non-stationarity, out-of-sample testing

1 Introduction

1.1 Motivation and thesis

Search "gamma exposure" and you will find a thousand charts of a jagged line called *GEX*, a vertical marker called the *gamma flip* or *zero gamma*, and a confident story: above the flip the market is calm and mean-reverting because dealers "sell strength and buy weakness"; below it the market is fast and trending because dealers do the opposite. The story is not wrong. It is a real mechanism, grounded in how options market makers manage inventory, and it has genuine support in the peer-reviewed literature. But almost every public presentation of it commits the same two sins: it treats an *estimate built on an assumption* as if it were a *measurement*, and it shows a historical GEX series that no freely available data can actually reconstruct.

The thesis of this paper is that the dealer-hedging mechanism is real and worth understanding precisely, and that the honest version of it is *smaller, more conditional, and more fragile* than the popular version. The single most consequential number in any GEX calculation — the sign of the dealer's position at each strike — is not observed. It is assumed. Change the assumption within a range that reasonable people defend, and the same option chain on the same day can tell you the market is in a positive-gamma regime, a negative-gamma regime, or exactly on the boundary. Most of the work of this paper is spent making that fact impossible to ignore, and then showing what can still be said once it is admitted.

Because almost no rigorous public treatment of this topic exists, the differentiation here is not cleverness; it is *honesty*. Where the data cannot support a claim, we say so and show why, rather than manufacture a plausible-looking series.

1.2 Scope, prerequisites, and intended reader

This is a research-grade paper in the Trading Library's *Options* track. From the greeks-and-hedging paper it takes as given: the Black-Scholes-Merton greeks and their signs; what it means to be *delta-neutral* and how a delta hedge is rebalanced; gamma as the curvature of delta, and the long-gamma/short-gamma asymmetry in a hedged book. That paper is a **mandatory** prerequisite. The volatility-surface paper (term structure, VIX/VVIX, the vol-of-vol complex) is **recommended** and is the natural companion for the vanna material in Section 7. The backtesting paper supplies the pre-registration and out-of-sample discipline used throughout and is the destination of Section 10's hand-off; it is recommended for the evidence sections.

The reader is assumed comfortable with a short derivation, a probability density, and a few lines of Python that pull a price series from a free feed. No measure theory and no stochastic calculus are required; the one or two integrals are done in words.

After reading this paper, a reader can:

- explain, without hand-waving, why market makers hedge delta mechanically and warehouse the rest, and who the "dealer" aggregate actually is in a 2025-era index-options market;

- state the *identification problem* precisely — separate what the option chain observes from what a GEX number assumes — and name the standard positioning assumptions, their provenance, and the two regimes in which they break;
- construct GEX from a raw option chain under an explicit sign convention, locate the gamma flip and the call/put walls, and quantify how far the flip moves when the estimator changes;
- distinguish the deterministic *charm* re-hedging flow from the volatility-driven *vanna* flow, and say when each dominates;
- read the 0DTE volume data for what it does to every GEX estimator, and to the very idea of a daily gamma number;
- and — most importantly — tell the difference between the mechanism as a *hypothesis with literature behind it* and the mechanism as a *confirmed result on one's own data*, which in this pipeline it is not yet.

1.3 Data and reproducibility

Nothing in this paper is called *validated* on the strength of external sources alone: a result drawn from the literature makes a mechanism *plausible*, never *confirmed*. Only an own reproducible result that survives a pre-registered out-of-sample test earns the stronger word, and — this matters for what follows — **no GEX-to-volatility relationship in this paper has yet earned it**. The mechanism is presented as a well-motivated, literature-backed hypothesis whose in-house confirmatory test is designed in Section 10 and not yet run.

Free-data pledge. Every number and figure in this paper uses only freely and publicly available data: the CBOE public delayed-quotes API for the live option chain, Yahoo Finance daily closes for SPX and VIX, and FRED (VIXCLS) as the documented alternative. Downloaded data are cached under `figures/data/` so each figure regenerates offline. This paper deliberately reproduces **no** proprietary vendor GEX levels, no licensed session readings, and no subscription-service screenshots; where the popular presentation of GEX relies on such a product, we rebuild the quantity from scratch on free data and report whatever it shows. Each figure caption ends with the script that reproduces it.

This is a research and educational document, not investment advice.

1.4 Organization of this paper

The paper is complete; every section below is written in full, and the arc is deliberate. Section 2 establishes the market maker's business and the inventory problem that makes delta-hedging mechanical. Section 3 — placed *before* any GEX formula, and given full weight — is the identification problem: what the chain observes versus what a positioning number assumes. Section 4 constructs GEX on a real free chain, compares sign conventions and three estimators explicitly, and locates the flip and the walls. Section 5 turns to price dynamics: the positive/negative-gamma regime story, an honestly labelled proxy regime study with an out-of-sample split, and a pre-committed pinning test. Sections 6 and 7 derive the two second-order re-hedging flows, charm (deterministic, time-driven) and vanna (volatility-driven). Section 8 is the 0DTE ecosystem and what it does to every earlier estimator. Section 9 is a catalogue of

what GEX *cannot* do — the failure modes and the competing explanations for the same stylized facts. Section 10 translates each mechanical claim into a pre-registered falsifiable test and describes the fully reproducible free-data pipeline end to end. The paper closes with annotated references.

2 The market maker's business

Before GEX can mean anything, one has to be clear about whose exposure it is supposed to measure and why that person is forced to trade the underlying at all. GEX is a statement about the hedging obligations of the aggregate options *dealer*. So we start with the dealer.

2.1 Quoting both sides and the inventory it creates

An options market maker's business is to quote a two-sided market — a bid and an offer — on a large set of strikes and expiries, continuously, and to earn the spread between them. The market maker does not, as a rule, have a view on where the index is going. What the market maker sells is *immediacy*: the willingness to take the other side of whatever a customer wants to do, right now, at a posted price.

The immediate consequence is *inventory risk*. If customers on balance buy the 7,600 put from the dealer, the dealer is now short that put — short a position with real delta, gamma, vega, and theta — not because the dealer wanted the exposure but because providing liquidity created it. Multiply this across thousands of strikes and expiries and the dealer holds a large, path-dependent book that it never chose on directional grounds. The entire discipline of dealer risk management is the management of that unwanted inventory.

2.2 Why delta is hedged mechanically and the rest is warehoused

Not all of the inherited risk is treated the same way, and the asymmetry is the seed of everything downstream.

- **Delta is hedged, mechanically and continuously.** Delta — sensitivity to the underlying — is the one exposure a market maker can neutralize cheaply and precisely, by trading the underlying (index futures, ETF shares, or the basket). It is also the exposure with the largest and fastest profit-and-loss impact. So the dealer runs the book close to *delta-neutral*, buying or selling the underlying whenever the net delta drifts outside a tolerance band. This hedging is not discretionary and not a market view; it is a near-automatic response to the book's changing delta. That mechanical response is the flow GEX is about. (Natenberg 2015; Garleanu, Pedersen & Poteshman 2009)
- **Gamma, vega, and theta are warehoused and priced.** The dealer cannot cheaply neutralize the curvature (gamma), the volatility exposure (vega), or the time decay (theta) of a large book — there is no single instrument that offsets them all without re-introducing the very optionality being hedged. So the dealer *warehouses* these, manages them within limits, lays some off against other customer flow, and is compensated through the bid-ask spread and the implied-volatility level at which it trades.

The crucial point for GEX is the interaction of the two. The dealer holds *gamma* it did not choose, and it hedges *delta* mechanically. Gamma is precisely the rate at which delta changes as the underlying moves.

So the size and sign of the warehoused gamma dictate the *direction and magnitude of the mechanical delta-hedging trades* the dealer must make as the market moves. A dealer who is *long gamma* buys the underlying as it falls and sells it as it rises — trading against the move. A dealer who is *short gamma* does the opposite — trading with the move. Section 5 is entirely about the market-level consequence of that single sentence; the rest of this paper is about measuring its inputs.

2.3 Gamma rent: why the sign of the inventory dictates the hedge

The long/short-gamma asymmetry is worth making quantitative, because it is the engine of every market-level claim in this paper. Take the simplest case: a dealer who has *sold* a customer one at-the-money put and hedged it to delta-neutral. A short put has *positive* delta, so the initial hedge is to *short* the underlying. Now let the underlying fall. The short put's delta rises toward +1, so the dealer's net delta drifts positive and, to restore neutrality, the dealer must *short more* underlying — it *sells into the falling market*. Let the underlying rise instead: the short put's delta falls toward zero, and the dealer *buys back* underlying — it *buys into the rising market*. The short-gamma dealer's hedge is *pro-cyclical* in both directions: it sells low and buys high.

Selling low and buying high has a name — it is a loss, paid continuously, and it is the *cost* of being short gamma. Over a rebalancing interval in which the underlying moves by ΔS , the hedging loss of a short-gamma position scales with $\frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$ — quadratic in the size of the move, which is why the pain of a short-gamma book grows non-linearly as volatility rises. (Natenberg 2015) The dealer is compensated for this by *theta*: the option it is short decays in its favour every day. The trade is therefore a bet that *realized* volatility will come in below the *implied* volatility at which the option was sold — the classic gamma-theta tradeoff. A dealer who is *long gamma* runs the mirror image: its rebalancing *buys low and sells high* (a profit, "gamma scalping"), paid for by the theta it bleeds, and it profits when realized volatility exceeds implied.

This is the microfoundation of the regime story, and it explains not just the *direction* of the hedging flow but why the flow is *stabilizing when long and destabilizing when short*. The long-gamma dealer is happy to keep leaning against moves because each lean is a profitable scalp; its hedging supplies liquidity and damps ranges as a by-product of making money. The short-gamma dealer chases moves because it must, and each chase is a loss; its hedging withdraws liquidity and amplifies moves, and — crucially — the loss accelerates precisely when volatility rises, which can force it to hedge *more aggressively* exactly when doing so hurts the tape most. The sign of the warehoused gamma is thus not a bookkeeping detail; it is the difference between a liquidity provider and a momentum accelerant. Everything after this section is an attempt to measure that sign from data that does not contain it.

2.4 Who "the dealer" actually is

The popular GEX chart speaks of "the dealer" as if it were one desk. It is not. In a 2025-era index-options market the aggregate liquidity-provider is a heterogeneous set of electronic market makers and proprietary trading firms, alongside bank derivatives desks that hedge structured products and the counterparties to systematic volatility-selling vehicles. These books differ in size, sign, horizon, and

hedging style. Some hedge in the E-mini future, some in ETF shares, some in the cash basket; some rebalance on a tight band, some on a schedule; some are net long gamma in one part of the surface and short in another.

Two consequences follow, and both are load-bearing for the honesty of everything after:

"The dealer" is a modelling fiction, and GEX is an aggregate estimate over it. There is no account whose gamma GEX reports. GEX sums an *assumed* dealer position across a heterogeneous industry, then treats the sum as if a single agent would hedge it in one instrument on one band. That is a useful abstraction and a genuine source of error, and it should be carried as a caveat, not buried.

The abstraction is useful because, in aggregate and on average, the mechanical delta-hedging of a large short-put/long-call book does push the same way. It is a source of error because the aggregation hides sign disagreement across desks, because hedging happens in several instruments with different frictions, and because in stress a capital-constrained desk may hedge *more* pro-cyclically than the frictionless story assumes.

Three frictions in particular separate the textbook flow from the executed one, and each widens the gap between a GEX number and the trades it is supposed to summarize. First, **the hedge is not continuous**. No desk rebalances on every tick; each holds a *tolerance band* around neutral and trades only when net delta breaches it. Between breaches there is no flow at all, and the same standing gamma produces very different executed volumes depending on how wide the bands are and how far price has travelled — which is why a large gamma *stock* need not imply any hedging *flow* on a quiet day (a distinction Section 9.2 makes central). Second, **the hedge is spread across instruments**. Some desks hedge SPX exposure in the E-mini future, some in SPY shares, some in the cash basket, each with its own cost, tick size, and liquidity; an aggregate GEX that imagines one agent hedging one instrument on one band is averaging over a genuinely heterogeneous execution. Third, **capital constraints make the frictionless story optimistic in exactly the wrong regime**. The frictionless account assumes the short-gamma dealer can always fund its accelerating hedge; in a stress event, margin and balance-sheet limits can force a desk to reduce risk *faster* than the smooth story predicts, so realized amplification below the flip can exceed the mechanical estimate. GEX is silent on all three, which is one more reason to read it as an upper-bound scenario rather than a forecast of flow.

2.5 The book is a shape, not a number

One last property of the dealer's inventory has to be stated before it is collapsed into a scalar, because the collapse throws it away. The warehoused gamma is not a single quantity; it is a *profile* over strikes and expiries. A dealer can be long gamma at the wings and short near the money, or long in the front month and short further out, and two books with the *same* total GEX can hedge in opposite directions depending on where spot happens to sit relative to their shapes. What actually governs the next hedging trade is the *local* gamma at the current spot and how fast that local gamma is changing as spot moves — the gradient of the profile, the "gamma of gamma." A large positive total can coexist with locally negative gamma right at spot, and it is the local sign that dictates whether the next move is damped or amplified.

This is why the single headline GEX number is the least informative representation of the book, and why the strike-level profile, the call and put walls, and the flip (Section 4.4) exist at all: they are attempts to communicate the *shape* that the scalar discards. Reading the grand total as though the whole book hedged uniformly at every price is the aggregation error in its purest form. The honest object of study is the profile and the position of spot within it, not the sum — a point the construction section takes as its organizing principle.

We now make the deeper version of this problem precise, before writing the formula it infects.

3 The identification problem

This section comes before any GEX formula on purpose. Every construction in Section 4 takes an option chain and returns a signed dealer-gamma number, and every such construction is only as good as one hidden step: the map from what the chain *shows* to who is *long or short*. That step is an assumption. Naming it, sourcing it, and stating where it fails is the difference between this paper and a screenshot with a red line on it.

3.1 What the chain observes and what it does not

Take a single option contract at a single strike and expiry. From free public data one observes, for that contract:

- **Open interest (OI)** — the number of contracts currently outstanding (opened and not yet closed or expired).
- **Volume** — the number of contracts that traded during the session.
- **Implied volatility and the greeks** — the market's IV and, from it, delta, gamma, and the rest, all computable from an option pricing model given the quote.
- **The last/settlement price and the quote.**

None of these says *who holds the contract long and who holds it short*. Every open contract has a long side and a short side by construction; OI counts the pair once. Volume records that a trade happened, not whether the customer was the *buyer* or the *seller* and therefore not whether the dealer ended up short or long the contract. Implied volatility is a price, not a position. So the object GEX most needs — the *sign of the dealer's net position at each strike* — is exactly the object the data does not contain. (this is a statement about the fields in the cached CBOE chain; see Section 4)

It is tempting to think trade *classification* can rescue the sign — that one could label each print a customer-buy or a customer-sell from its position relative to the bid and offer, as equity-microstructure work does with the Lee-Ready rule, and net them into an inferred dealer position. This fails for options for three compounding reasons. Signed order flow decays: a contract opened as a customer buy a month ago may since have been sold, rolled, or offset, and the standing open interest carries no memory of the sequence that built it. Multi-leg structure hides direction: a large share of options volume trades as spreads, combinations, and delta-hedged packages in which the "buy" and "sell" legs are booked

together, so the sign of any single leg's print says little about the customer's economic position. And the dealer is not the only non-customer: other market makers, arbitrageurs, and each other's hedges populate the tape, so a print at the offer is not reliably a customer lifting a dealer. Even with a full signed-trade feed — which the free pipeline does not have — reconstructing the *net* dealer inventory at each strike from prints is a fraught inference, not a lookup. The honest conclusion is that the sign is *modelled*, and the only question is how transparent the model is.

The identification problem, in one line. The chain tells you *what exists and what it is worth*; it does not tell you *who is on which side*. GEX needs the second and only measures the first, so it must supply the missing sign from an assumption.

3.2 The standard assumption set and where it comes from

The dominant convention — popularised by the SqueezeMetrics *Gamma Exposure* white paper and reused, often silently, by nearly every public GEX product — infers dealer positioning from a stylised picture of *customer* flow in index options: (SqueezeMetrics 2017)

- **Customers are net sellers of index calls** (systematic call *overwriting* by institutions and yield-seeking funds), so the dealer is net *long* calls, hence *long gamma* from the call book.
- **Customers are net buyers of index puts** (portfolio *protection* and hedging), so the dealer is net *short* puts, hence *short gamma* from the put book.

Under this picture the standard sign convention is *calls positive, puts negative* in the gamma sum. The provenance is important: this is a faithful description of the *pre-2020 institutional index-options world*, in which overwriting and protective-put programs dominated identifiable flow. It was a reasonable modal assumption for SPX in that era, and there is real economic content to it — the demand-based option-pricing literature shows that persistent net customer demand does move prices and implied volatilities in exactly the direction these programs imply. (Garleanu, Pedersen & Poteshman 2009)

The provenance is worth dwelling on, because it explains both the assumption's plausibility and its expiry date. The picture is a lineal descendant of the portfolio-insurance world that predates listed index options and hardened after 1987: large asset owners are structurally *long* the market and structurally *short* volatility protection, so their derivatives footprint is dominated by buying downside puts and selling upside calls to finance them. Bank structured-product desks reinforce the same asymmetry — the retail products they hedge typically leave the desk long downside and short upside vega. So the standard convention is not arbitrary; it encodes a genuine, decades-long institutional demand asymmetry. But that also fixes its shelf life: the convention is only as good as the persistence of the demand asymmetry it encodes, and it says nothing about a *new* population — retail call buyers, systematic 0DTE premium sellers — whose footprint the original picture never contemplated.

But it is, in the end, an assumption about the *typical* customer, frozen from a particular decade, and it is applied uniformly across every strike regardless of what is actually trading there. That is the crack.

3.3 What breaks: the 2021 meme regime and the 0DTE era

Two structural shifts since 2020 attack the standard assumption at its most sensitive input, the sign of the call book.

- **The 2021 meme regime.** A wave of retail activity in short-dated, out-of-the-money *call buying* inverted the assumed direction of call flow in the names it touched. If customers are net *buying* calls, the dealer is net *short* calls — *short gamma* where the convention assumes long gamma. The sign of an entire book flips. Even in index products the episode was a proof of concept that the "customers overwrite calls" premise is a historical contingency, not a law.
- **The 0DTE era.** Since daily SPX expirations arrived in 2022, a large and growing share of activity is same-day, two-sided, and dominated by strategies (short-dated credit spreads, iron condors, and outright directional punts) whose net dealer-positioning sign is genuinely ambiguous and, worse, *intraday*. Much 0DTE flow is customers *selling* premium, which would make dealers *long* both calls and puts and hence long gamma — the opposite of the protective-put picture — while much is customers *buying* lottery-ticket directional options, which makes dealers short. The aggregate sign is not stable within a single day, let alone assumable from a fixed rule. Section 8 quantifies just how much of the volume this is. (CBOE 2023)

The two shifts compound rather than cancel, and the data make the compounding concrete. Section 8 will show that on the cached snapshot the same-day book is about 59% of the day's volume but only 1.4% of the end-of-day open interest a GEX build actually sees. So the block of contracts whose *sign* is most ambiguous (two-sided 0DTE flow) is also the block that most dominates the session's hedging *and* the block that least survives into the open-interest snapshot the estimator reads. The standard convention, calibrated to a slow institutional book, is being stretched over exactly the fast, retail-heavy contracts it was never designed to describe — and then evaluated on an OI snapshot from which most of those contracts have already vanished.

The consequence is not that GEX is useless; it is that *the most important input to GEX is least reliable exactly when GEX is most watched* — in fast, retail-heavy, short-dated conditions. A number whose sign depends on an assumption that was calibrated to a bygone flow regime should be read as a scenario, not a reading. Section 4 shows this concretely: on one real chain, three defensible ways of supplying the missing sign put the regime boundary on three different sides of spot.

3.4 A worked ambiguity at a single strike

To make the problem tangible rather than abstract, follow one strike loaded with same-day open interest — the kind of near-the-money 0DTE strike that dominates a session's hedging. The chain shows its open interest and the day's volume; both are large. Now ask the only question that matters for GEX: were the customers who built that position net *sellers* of premium or net *buyers*? If the flow was customers selling credit spreads and iron condors — a common 0DTE pattern — then the dealer is *long* that gamma, and the strike contributes with a positive sign, hedging anti-cyclically. If the flow was customers buying short-dated directional options, the dealer is *short* the same gamma, and the strike contributes with the

opposite sign, hedging pro-cyclically. The observable fields — strike, expiry, open interest, volume, implied volatility, gamma — are *identical* in the two worlds. Only the unobserved sign differs.

The magnitude of the swing is not marginal. A near-the-money 0DTE strike can carry a dollar-gamma contribution of several hundred million per 1% move (Section 4.1); flipping its sign is therefore a swing of *twice* that — potentially more than half a billion dollars of hedging direction — at a *single* strike, decided by an input the data does not contain. Aggregate a chain full of such strikes, each with its own hidden sign, and you regenerate exactly the estimator disagreement Section 4.3 measures: not because the arithmetic is uncertain, but because the signs are assumed. This is the identification problem made concrete, and it is why the next section spends its effort on sensitivity rather than on a single confident number.

4 GEX construction

With the identification problem stated, we can build the number honestly — meaning: with every assumption on the table and with the sensitivity of the answer to those assumptions shown, not hidden. All computations in this section use a single cached CBOE delayed-quotes snapshot of the SPX chain (17 September 2026, spot 7,637.76, 29,416 live contracts after excluding the same-day settled expiry).

4.1 The naive spot-gamma sum and its scaling

The building block is *dollar gamma*: how much a contract's delta changes for a move in the underlying, converted to money. For a single option with per-contract gamma Γ (the pricing-model gamma, delta change per one-point move in the underlying), open interest OI , and contract multiplier M (100 for SPX), the change in the position's aggregate delta for a small underlying move is $\Gamma \cdot OI \cdot M$. Multiplying by the underlying level S converts a *one-point* move to the dollar delta that must be re-hedged, and multiplying by S again — with a factor of 0.01 — expresses the hedging requirement per *one-percent* move in the index:

$$GEX(i) = s_i \cdot \Gamma_i \cdot OI_i \cdot M \cdot S^2 \cdot 0.01,$$

where $s_i \in \{+1, -1\}$ is the dealer-position sign supplied by the positioning assumption of Section 3. Summing $GEX(i)$ over all live contracts gives aggregate dealer GEX in dollars of hedging per 1% move. The two scaling choices — dollar gamma per one-point move ($\Gamma \cdot OI \cdot M \cdot S$) versus dollar gamma per one-percent move (the $S^2 \cdot 0.01$ form) — are both in public use and differ only by a factor of $S/100$; we fix the per-1% form throughout so that levels are comparable across time and instruments. This is a units convention, not an assumption about positioning.

To read the units concretely, take one strike, illustratively, with 5,000 contracts of open interest and a per-contract gamma of 0.001, at a spot of 7,600. Its contribution is $0.001 \cdot 5,000 \cdot 100 \cdot 7,600^2 \cdot 0.01 \approx \0.29 billion per 1% move. The interpretation is operational: *if* dealers are net long that strike and the index moves 1%, roughly \$0.29 billion of underlying must be traded to keep the book delta-neutral, in the anti-cyclical direction; if net short, the same size trades pro-cyclically. Summed across the tens of thousands of live contracts in a real chain, the aggregate lands in the single-digit billions per 1% (Figure 2 sums the

CBOE-supplied gamma across the chain to +\$5.4 billion per 1% for this snapshot). That headline sign is deceptively firm: recomputing the very same book's gamma from Black–Scholes and evaluating it at spot lands at −\$28 billion per 1% (estimator E1 in Section 4.3) — the *opposite* sign — because this snapshot sits almost exactly on the gamma flip (+0.24% above spot), where the net is a knife-edge difference between huge, near-cancelling call and put exposures. The number is the size of the mechanical hedging demand a 1% move *would* generate — a potential, not a realized flow, and, as that sign disagreement already warns, only as reliable as the sign in front of it.

Two things in this formula are *observed* (Γ , OI) and one is *assumed* (the sign s_i). All of the fragility lives in the sign.

4.2 Sign conventions compared explicitly

Two sign conventions dominate, and they are not close:

- **Standard convention** (calls +1, puts −1): the SqueezeMetrics picture of Section 3.2 — dealers long the call book, short the put book. This is the convention behind Figure 2.
- **"Customers long everything"** (calls −1, puts −1): the naive alternative in which every option is assumed customer-held and every dealer position short. Under this convention the dealer is short gamma everywhere and aggregate GEX has no sign change at all — there is no gamma flip, because there is no regime boundary to cross.

These are not two flavours of the same number; they are two different theories of who holds the book, and only one of them (at most) can describe a given day. The standard convention at least *allows* a regime boundary to exist. Whether it locates it correctly is an empirical question the free data cannot settle, because — per Section 3 — the free data does not contain the sign.

4.3 One chain, three estimators, three flip points

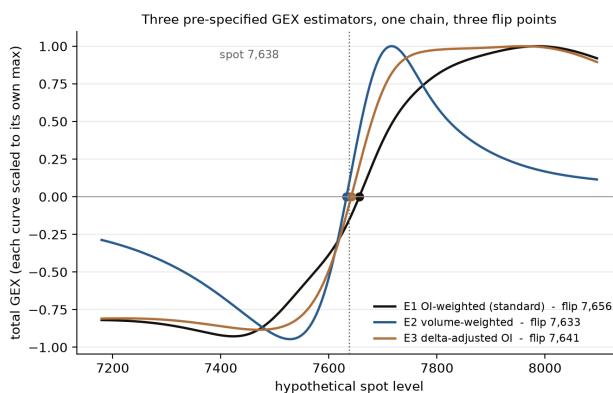
To make the sensitivity concrete, we hold the chain, the pricing model, and the scaling fixed and vary *only* how the observable book is turned into a weighted dealer position. Three estimators, all fully specified in advance, no tuning:

- **E1, OI-weighted (standard):** weight $w_i = \text{OI}_i$, signs calls +, puts −. The textbook GEX.
- **E2, volume-weighted:** weight $w_i = \text{volume}_i$, signs calls +, puts −. Uses the day's *flow* rather than the standing book, on the view that recently traded contracts are the ones being actively hedged.
- **E3, delta-adjusted OI:** weight $w_i = \text{OI}_i \cdot (1 - |\delta_i|)$, signs calls +, puts −. Discounts deep-in-the-money contracts, because the canonical customer-flow stories (overwriting, protective puts) live at and out of the money; one transparent member of a family, not an endorsement.

Total GEX for each estimator is re-evaluated across a grid of hypothetical spot levels using Black–Scholes–Merton gamma at each contract's own implied volatility, and the zero crossing of that curve is the estimator's *gamma flip*. The results, on the one snapshot:

Estimator		Positioning weight	Sign convention	Net GEX at spot (\$bn/1%)	Gamma flip	Flip vs spot
E1	OI-weighted (standard)	OI	calls +, puts -	-28.2	7,656	+0.24%
E2	volume-weighted	volume	calls +, puts -	+7.9	7,633	-0.07%
E3	delta-adjusted OI	$OI \cdot (1 - \delta)$	calls +, puts -	-11.2	7,641	+0.05%
E4	customers long everything	OI	calls -, puts -	-750.7	none	—

Read the table slowly, because it is the empirical heart of the paper. The three standard-sign estimators put the flip within a band of roughly 23 index points — about a third of a percent of spot — which sounds like agreement. It is not. The snapshot spot sits *inside* that band, so the three estimators disagree on the thing that actually matters: the *sign of the regime at the current price*. E1 says net dealer gamma at spot is negative (-28 \$bn/1%), i.e. a fragile, amplifying regime; E2 says it is positive (+8 \$bn/1%), a damping regime; E3 says mildly negative. The fourth convention removes the flip entirely. Same chain, same day, same model — opposite regime calls (Figure 1).



CBOE delayed quotes 2026-09-17, 28293 live contracts; same-day expiry excluded; BSM gamma at per-contract IV, $r=0.04$, $q=0.01$. Curves scaled to unit max for comparability; flips are scale-invariant.

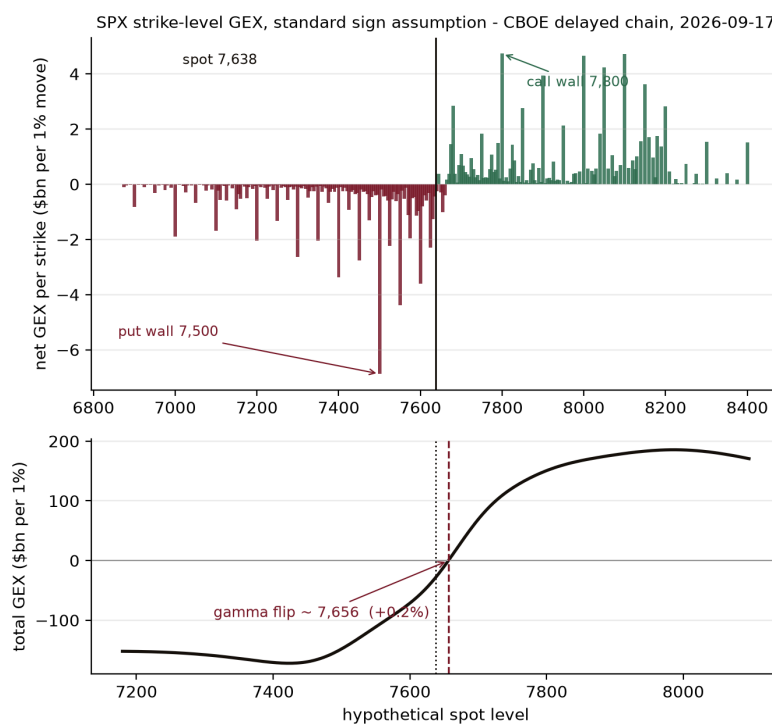
Figure 1. Three GEX estimators applied to one SPX chain (CBOE delayed quotes, 2026-09-17, spot 7,638), differing only in how the observable book is turned into a dealer position: OI-weighted (flip 7,656, +0.24%), volume-weighted (flip 7,633, -0.07%), and delta-adjusted OI (flip 7,641, +0.05%). The three flip points span about 23 index points and the sign of net dealer gamma at spot disagrees across them (-28, +8 and -11 dollar-billion per 1% move); a fourth convention, customers long everything, has no zero crossing at all. Reproduce with `figures/fig_gex_estimators.py`.

4.4 The strike-level profile: flip and walls

Fixing the standard convention (E1), the strike-level profile shows where the gamma sits. Figure 2 plots net GEX per strike and the total re-evaluated across spot. The largest positive-gamma strike — the *call wall*, where dealer long-gamma is concentrated and hedging most strongly resists an upward push — is at 7,800; the largest negative-gamma strike — the *put wall* — is at 7,500. The total curve crosses zero at the

gamma flip near 7,656, which on this snapshot is +0.24% *above* spot, meaning the standard estimator places the market a hair into negative-gamma territory.

A methodological check worth stating, together with its limits: the exposure figure can be built with the CBOE-supplied gamma directly, or with Black-Scholes gamma recomputed from each contract's implied volatility. Near the money the two agree closely (median ratio 1.001, computed on $|K/S - 1| < 0.02$) — but that near-the-money median says nothing about the wings, where the large, near-cancelling exposures that decide the *net* actually live. Two further caveats matter for the headline numbers. The +\$5.4 bn figure sums the CBOE gamma across *all* live contracts, whereas the -\$28 bn estimator E1 is built on the filtered set live & (iv > 0.01) & (oi > 0) and prices time as $\max(\text{dte}, 1)/365$, which forces 1-DTE contracts onto a calendar-year fraction: the two are *different populations under different time conventions*, not one book measured through two gamma sources. So the sign difference is partly the knife-edge flip and partly those construction choices — which is all the more reason not to wave the gamma input away as settled. The decisive uncertainty is still the *sign*, exactly as Section 3 predicted; the honest statement is that both the sign and the exact magnitude of the net depend on modelling choices the free data does not pin down.



CBOE delayed quotes 2026-09-17; 29416 live contracts, same-day expiry excluded from exposure; signs: calls +, puts - (standard assumption); $r=0.04$, $q=0.01$.

Figure 2. SPX strike-level gamma exposure from the free CBOE delayed chain (2026-09-17, spot 7,638, 29,416 live contracts) under the standard sign convention (dealers long calls, short puts). Top: net GEX per strike, with a call wall at 7,800 and a put wall at 7,500. Bottom: total GEX re-evaluated across spot, crossing zero — the gamma flip — near 7,656, about +0.24% above spot, so the snapshot sits almost exactly on the regime boundary.

Recomputed Black-Scholes gamma matches the CBOE-supplied gamma to within 0.1% near the money.

Reproduce with `figures/fig_gex_profile.py`.

4.5 Aggregation choices: net versus gross, and which expiries count

Two further choices sit between a chain and a GEX number, and both are frequently left implicit. The first is **net versus gross**. GEX as defined here is a *net* signed sum — it answers "which way does the aggregate hedge push?" A different, equally common product sums the *absolute* dollar gamma across strikes, discarding the sign — it answers "how much hedging turnover will a move create?" These are answers to different questions, and a chart that does not say which it plots invites the flow-versus-stock confusion of Section 9.2: a large *gross* gamma near expiration reliably implies large hedging *volume*, while only the *net* gamma speaks to a directional damping or amplification. Reporting one and interpreting it as the other is a category error.

The second is **which expiries enter the sum**. Per-contract gamma is largest for near-the-money, near-expiry options and vanishes for far-dated ones; open interest runs the other way, concentrated far out on the curve (Section 8 shows 53% of OI beyond 63 days on this snapshot, against 1.4% at zero days). The net sum is therefore dominated by the near-term near-the-money book, but the *choice* of expiry window still moves the number materially: a "total GEX" that includes every listed expiry, a "front-month GEX," and a "0–2 day GEX" are three different series with three different flip points, and the popular single line rarely states which it is. Combining SPX and SPY exposures compounds the problem, since the two share a 100 multiplier but differ tenfold in notional; naive addition double-counts or mis-weights unless converted to a common notional first. None of these choices involves the positioning sign — they are all *construction* decisions — yet each shifts the headline, which is why a reproducible pipeline (Section 10) must pin them down in advance rather than inherit a vendor's silent defaults.

4.6 What the sensitivity means

The estimator table is not a curiosity; it is the operational content of the identification problem. A flip that moves 23 points and flips sign at spot depending on a defensible modelling choice cannot be read as a precise level. Two disciplines follow. First, any use of the flip should be as a *zone with a sign that may be wrong*, not a line. Second, the distance between spot and the flip — small here — is itself information: when spot sits on the flip, the regime call is least robust, and honest practice is to widen the uncertainty rather than pick the estimator that gives the cleaner story. The zero-gamma level tells you the *mode* the mechanism would produce; it does not, on free data, tell you which side you are on with confidence.

A final construction subtlety separates two things the public vocabulary conflates under "zero gamma." The *classical* zero-gamma level is a parity condition on the chain itself — the price at which aggregate call gamma equals aggregate put gamma — and it is computed from open interest and gammas with *no* dealer-positioning sign required. The *measured* gamma flip used above is the price at which the signed *dealer* gamma crosses zero, which does require the sign convention. These are not the same level, and they need not even move together: the classical parity level is a property of the OI distribution, while the measured flip depends on how that distribution is signed and weighted. Products that estimate the *dealer's* flip from richer positioning models tend to place it *earlier* than the classical parity level and to update it intraday, so a chart labelled "zero gamma" can mean either object depending on the vendor. For a free-data pipeline the honest move is to compute the classical, sign-free parity level as the reproducible

anchor, report the measured flip only alongside its explicit sign convention, and never present either as the single true regime boundary.

5 Gamma regimes and price dynamics

Section 2 established the mechanism at the level of a single dealer; this section states its market-level consequence and then tests, as honestly as the free data allow, whether the consequence shows up in prices.

5.1 The mechanism: damping above, amplifying below

The consequence follows directly from Section 2.2. When the aggregate dealer is *net long gamma* (positive GEX, spot above the flip), delta-hedging is *anti-cyclical*: as the index rises the book's delta grows and the dealer *sells* the underlying; as it falls the dealer *buys*. The hedging flow leans against the move, supplying liquidity, compressing ranges, and pulling price back toward where the gamma is concentrated — a damping, mean-reverting regime. (Barbon & Buraschi 2021; SqueezeMetrics 2017)

When the aggregate dealer is *net short gamma* (negative GEX, spot below the flip), the same hedging logic runs the other way: the dealer *buys* into strength and *sells* into weakness, *pro-cyclically*, removing liquidity and amplifying moves. Because the required hedge grows as the move grows, the effect is nonlinear and can feed back on itself — the mechanism behind the "gamma fragility" and "gamma squeeze" language. (Barbon & Buraschi 2021)

The feedback is worth stating as a loop, because its shape explains both the danger and its limits. In a negative-gamma regime a down move forces the dealer to sell, which extends the down move, which enlarges the dealer's short delta, which forces more selling — a positive feedback that can compress a normal pullback into a fast, illiquid air-pocket. But the loop is *self-limiting* in two ways that keep it from being a perpetual-motion crash. First, moving through strikes changes the gamma profile itself: as spot travels, it can pass the flip and enter a positive-gamma region where the hedging reverses and damps, so the very move the loop produces can carry price into the zone that arrests it. Second, expiration and re-hedging retire the gamma that drove the episode; the 0DTE book that amplified an afternoon is gone by the close. This is why the mechanism produces *episodes* — sharp, bounded, often intraday — rather than a secular instability, and why its cleanest empirical signature (Barbon & Buraschi) is *intraday* momentum conditional on illiquidity, not a persistent drift. The honest reading is that negative gamma is a *fragility amplifier*, not a directional forecast: it makes the tape more sensitive to whatever shock arrives, without predicting the shock.

The mechanism is also *asymmetric*, in a way that matters for how it is tested. Under the standard convention, the dealer's short gamma is concentrated in the *put* book (protective-put demand), so the amplification channel is loaded on the *downside*: negative-gamma effects tend to show up most on down moves, while the positive-gamma damping is most visible in calm upward drifts. This downside-loading is observationally close to the *leverage effect* — volatility rising as prices fall — which is precisely why Section 9.4 insists the two be separated with controls rather than assumed distinct. It has a testing

consequence too: a symmetric test that treats up and down moves alike can dilute a real effect that lives predominantly on one side, so a well-designed regime test should allow the coefficient to differ by direction rather than impose symmetry it has no reason to expect.

The peer-reviewed anchor is worth stating precisely, because it is stronger than the vendor literature and more conservative than the blog version. Barbon and Buraschi document, in a large equity-options panel, that intraday *momentum* is associated with negative ex-ante dealer gamma imbalance and intraday *reversal* with positive imbalance, and that the effect is stronger where the underlying is *illiquid* — exactly the sign pattern the mechanism predicts, and conditional in exactly the way the mechanism requires. A related line finds that option-market delta-hedging accounts for a non-trivial share — on the order of a tenth — of daily absolute stock returns in their sample. (Ni, Pearson, Poteshman & White 2021) These are single-stock results; their transfer to the cash-settled, deeply liquid index is a hypothesis, not a finding.

The single most-quoted public number in support of the regime story deserves an explicit, critical airing, because most readers will have met it and few will have met its caveats. The SqueezeMetrics *Gamma Exposure* white paper reports that, sorting SPX days by its GEX estimate over 2004–2017, the top-GEX quartile realized a daily standard deviation of roughly 0.55%, against roughly 0.85% for the middle quartiles — a monotone fall in realized volatility as GEX rises. (SqueezeMetrics 2017) Taken at face value this is a striking, correctly-signed confirmation of the damping story. But four cautions must travel with it, and this paper attaches them rather than repeat the number bare. It is a *vendor* white paper, not peer-reviewed. It is *in-sample*: the relationship is reported on the same data used to build and present the estimator, with no out-of-sample split. It *ends in 2017*, before daily expirations and the 0DTE regime that Section 8 shows has changed the microstructure the number was measured in. And, most importantly, it is *not identified*: sorting on GEX is close to sorting on the volatility level itself (Section 9.3), so the fall in realized volatility across GEX quartiles is exactly what volatility clustering alone would produce. The number is a motivation for a test, not the test — and it is precisely the kind of correctly-signed, in-sample, level-confounded result this paper's method is built to distrust.

Direction is the whole claim. The regime story is a claim about the *sign* of a conditional relationship: positive gamma → damped, negative gamma → amplified. An effect of the right magnitude but the wrong sign refutes it. This paper's own research history contains a case where a gamma claim was recorded as confirmed by evidence that in fact showed the opposite sign; the discipline that catches it is a reflex direction check on every piece of evidence, and Section 9 returns to it.

5.2 Why the historical test cannot be run here, and what we do instead

To test the regime story properly one needs a *historical GEX time series*: for every past day, the strike-level open interest of the whole chain, from which dealer gamma is rebuilt under a fixed convention and merged with next-day realized volatility. Free public data does not contain historical strike-level OI. The CBOE delayed-quotes API gives a *live* snapshot only; Yahoo and FRED give prices and index-level VIX, not the chain's OI history. A real historical study requires an end-of-day options dataset with per-strike OI — for example optionsDX EOD files — which is outside this paper's free-data pledge.

It is worth being precise about what such a study would actually require, both so the gap is honest and so the work is well-defined for anyone who closes it. The minimum is a per-day, end-of-day snapshot of the *whole* SPX chain carrying, for each strike and expiry, the open interest, the volume, and enough to recompute gamma (implied volatility, or the vendor greek), over a span long enough to contain both regimes — a decade or more. optionsDX-style EOD files provide exactly this, in bulk, for a fee; that fee is the only reason it falls outside "free," not any technical obstacle. Given the history, the pipeline of Section 10 rebuilds a daily GEX series under one fixed sign convention and merges it with the free SPX and VIX daily data. But two honest caveats survive the upgrade. First, an end-of-day series inherits the 0DTE measurement gap of Section 8 — it is a series of the *slow* book, and a faithful 0DTE study needs *intraday* timestamped OI, which is rarer and costlier still. Second, and more fundamental, adding OI history solves the *time-series* problem (you finally have a past to test against) but does **not** solve the *identification* problem (you still assume the sign at every strike). The real study is therefore a genuine advance and still a conditional one: it can test whether the *assumed-sign* GEX predicts realized volatility out-of-sample, which is worth knowing, but it cannot certify the assumption itself.

We therefore refuse the most common shortcut, which is to display a GEX-like historical line reconstructed from something the data does not support, and instead do two smaller, honest things: a clearly labelled *proxy* regime study, and a *pre-committed* pinning test. Neither is a GEX study, and we never label them as one.

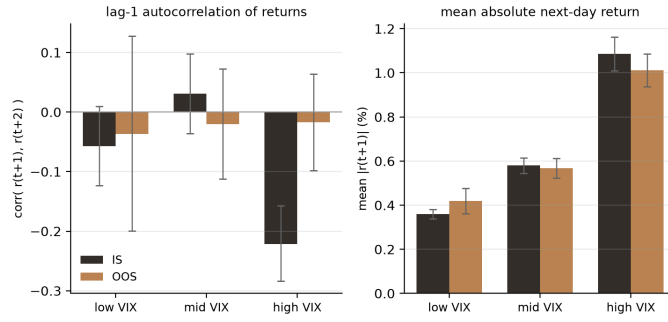
5.3 A proxy regime study — and why it is only a proxy

Because dealer gamma cannot be rebuilt historically here, we condition on the *VIX level* instead — an observable, free variable that correlates with the gamma regime (low VIX days tend to be positive-gamma, high-VIX days negative-gamma) but does *not* identify it. The design is pre-committed and reported in full, whatever it shows:

- **Data:** SPX daily closes (Yahoo ^GSPC) and VIX daily closes (Yahoo ^VIX; FRED VIXCLS is the documented alternative), 2011-09-19 to 2026-09-17, 3,771 sessions.
- **Regime proxy:** terciles of the prior-day VIX close. The tercile breakpoints (13.8 and 17.8) are estimated on the **in-sample** period only ($\leq 2021-12-31$) and applied unchanged **out-of-sample** ($\geq 2022-01-01$).
- **Metrics, both strictly after the conditioning day:** mean absolute next-day return $|r(t+1)|$; and lag-1 return autocorrelation, the correlation of $(r(t+1), r(t+2))$ across days in the cell. 95% confidence intervals: normal approximation for the mean, Fisher-z for the correlation.

The results (Figure 3) split cleanly into a robust part and a fragile part. The *magnitude* channel is robust: mean absolute next-day return rises monotonically with the VIX tercile in both samples — 0.36%, 0.58%, 1.09% in-sample and 0.42%, 0.57%, 1.01% out-of-sample. The *autocorrelation* channel is not: in-sample the high-VIX cell shows strongly negative lag-1 autocorrelation (-0.22 , CI $[-0.28, -0.16]$, excluding zero), consistent with the whipsaw/mean-reversion one would expect if a stressed, fast regime were present; out-of-sample that same cell collapses to -0.02 (CI $[-0.10, +0.06]$, including zero). The signature that looked like a regime effect in-sample does not replicate.

PROXY for regime conditioning: SPX daily dynamics by prior-day VIX tercile (this is not a GEX series)



Yahoo ^GSPC x ^VIX, 2011-09-19 to 2026-09-17; in-sample <= 2021-12-31 (tercile breakpoints 13.8 / 17.8 estimated in-sample only), out-of-sample 2022-2026. Error bars: 95% CIs (Fisher z / normal).

Figure 3. Proxy for regime conditioning (not a GEX series): SPX daily dynamics sorted by prior-day VIX tercile, breakpoints 13.8 and 17.8 estimated in-sample only and applied unchanged out-of-sample, 2011–2026, 3,771 sessions. Right: mean absolute next-day return rises monotonically with the VIX tercile in both samples (0.36 / 0.58 / 1.09% in-sample; 0.42 / 0.57 / 1.01% out-of-sample). Left: the lag-1 return autocorrelation is strongly negative in the high-VIX cell in-sample (−0.22, CI excludes zero) but collapses to −0.02 (CI includes zero) out-of-sample. Reproduce with figures/fig_regime_proxy.py.

Two caveats are mandatory, and they are why this can never be labelled a GEX result. First, **endogeneity**: conditioning on VIX to predict absolute return is close to conditioning on volatility to predict volatility. VIX is a forward volatility expectation; realized volatility clusters; so the monotone magnitude channel is largely the *volatility-clustering* stylized fact re-expressed, not evidence that dealer gamma damps or amplifies anything. A true GEX study must condition on the *gamma sign*, not on the vol level, precisely to break this circularity. Second, **non-stationarity**: the one channel that carried a regime-like signature (autocorrelation) is in-sample only, which is exactly the pattern one expects when a relationship is regime-dependent and the regime drifts — the 0DTE era changed the microstructure between the two samples. The proxy is consistent with the mechanism's *sign* on the magnitude axis and silent-to-negative on the dynamics axis, and it is not a substitute for the real test designed in Section 10.

5.4 A pre-committed pinning test

The damping story makes a sharp, falsifiable prediction near expiration: if dealers are long gamma at the big strikes, anti-cyclical hedging should pull the underlying *toward* those strikes into the close on expiration day — *pinning*. The literature supports pinning strongly, but with a crucial qualification: it is documented for *single stocks with physically deliverable shares*, where the mechanism has a concrete settlement channel, and it is conditional on the dealer gamma sign. (Ni, Pearson & Poteshman 2005; Golez & Jackwerth 2012) The cash-settled index cannot be held to expiry, so the mechanism is far weaker there, if present at all.

The mechanism in the single-stock case is concrete and worth stating, because it is what does *not* transfer. Ni, Pearson and Poteshman show that optionable stocks close disproportionately *at* strikes with large open interest on expiration Fridays, that this clustering is *absent* in the same stocks before they were optioned and in non-optionable peers, and that it is driven partly by the delta-hedge rebalancing of long-gamma option writers and partly by expiration-day trading tied to the physical settlement of in-the-

money options. (Ni, Pearson & Poteshman 2005) The channel needs a deliverable share and a settlement mechanic that a cash-settled index does not have. Golez and Jackwerth sharpen the point for the index complex: they find pinning in the S&P 500 *futures* — which are deliverable into the settlement — but conditional on the *sign* of dealer gamma, with pinning under long gamma and *anti*-pinning under short gamma. (Golez & Jackwerth 2012) That conditionality is the reason a single unconditional test on the cash index is expected to average toward no effect, and it is why the null in Figure 4 is the *predicted* outcome rather than a failure of the test. Reading the null correctly — as consistent with a real but conditional and venue-specific mechanism — is itself a small lesson in not over-claiming from a suggestive story.

We test it with one rule fixed in advance — no grid search, no threshold tuning: the distance of the SPX daily close to the nearest 25-point strike, on monthly-OPEX Fridays (the third Friday of the month) versus all other Fridays, 2011–2026. Under no pinning the distance is uniform on $[0, 12.5]$ with expected value 6.25 index points; pinning would push the OPEX-Friday mean *below* that and below the control. The pre-committed directional test is a one-sided Welch z on the two means. The result (Figure 4) is a clean null: OPEX Fridays average 6.28 points from the nearest strike, other Fridays 6.39 — both sitting essentially on the 6.25 no-pinning expectation, a difference of -0.11 points with one-sided $z = -0.37$ and $p = 0.36$. There is no detectable pinning of the *index* close to round strikes, exactly as the single-stock-deliverable reading of the literature predicts. Reporting this null is the point: it is what an honest pre-committed test looks like when the mechanism does not transfer.

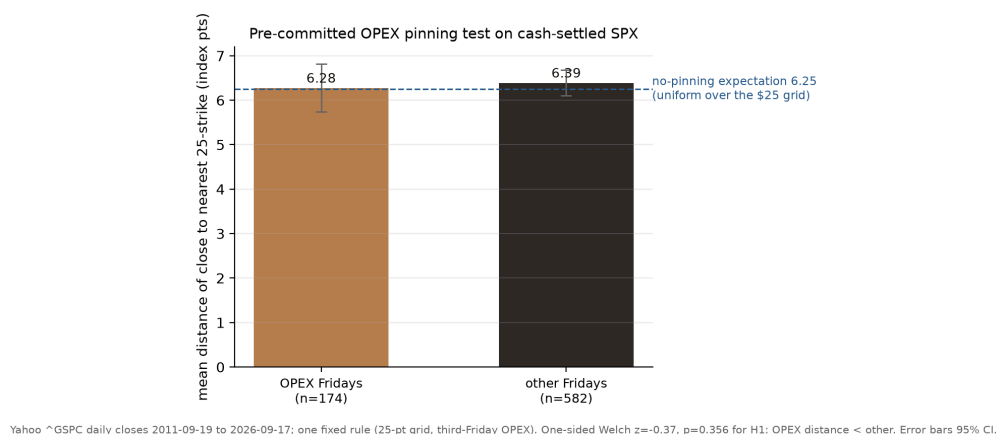


Figure 4. Pre-committed OPEX pinning test on cash-settled SPX, one rule fixed in advance: mean distance of the daily close to the nearest 25-point strike on monthly-OPEX Fridays ($n=174$) versus all other Fridays ($n=582$), Yahoo ^GSPC 2011–2026. Both means sit on the 6.25-point no-pinning expectation (6.28 vs 6.39); the OPEX-minus-other difference is -0.11 points, one-sided Welch $z = -0.37$, $p = 0.36$ — no detectable pinning of the index close. Reproduce with figures/fig_pinning_test.py.

5.5 Base rates and non-stationarity

The proxy and the pinning null together teach two disciplines the popular GEX literature almost never applies, and both are worth stating explicitly because they govern how any future GEX result must be read. The first is the *base rate*. The monotone rise of mean absolute return across VIX terciles is a base

rate — high-volatility days are, tautologically, more volatile — and it is the benchmark any genuine GEX signal must beat *incrementally*. A GEX quartile study that merely reproduces this monotonicity has demonstrated nothing beyond the base rate; the burden is to show information *after* the volatility level is already known, which is exactly why the Section 10 pre-registration puts lagged realized volatility and VIX in the control set before the GEX coefficient is allowed to claim anything.

The second is *non-stationarity*, and the proxy makes it visible. The lag-1 autocorrelation signature that was strong in-sample collapsed out-of-sample, and the collapse is not noise to be averaged away — it is information. It coincides with the 2022 arrival of daily expirations and the 0DTE regime, which changed the very microstructure the earlier relationship was measured in. A relationship estimated across such a boundary is not one relationship; it is a blend of two regimes, and its full-sample estimate describes neither. Two consequences follow for inference. Cells must be reported with their sizes and small ones (a rule of thumb: fewer than fifteen regime episodes) treated as unreliable regardless of a flattering point estimate. And because volatility clusters, the *effective* number of independent regime episodes is far smaller than the raw day count, so naive confidence intervals — including the ones drawn in Figure 3 — overstate precision; the correct inference uses block-bootstrap or Newey-West standard errors that respect the autocorrelation, as the pre-registration specifies. The proxy is honest about its own limits precisely by showing a signature that does not survive the regime change, which is the most useful thing a proxy can do.

6 Charm flows

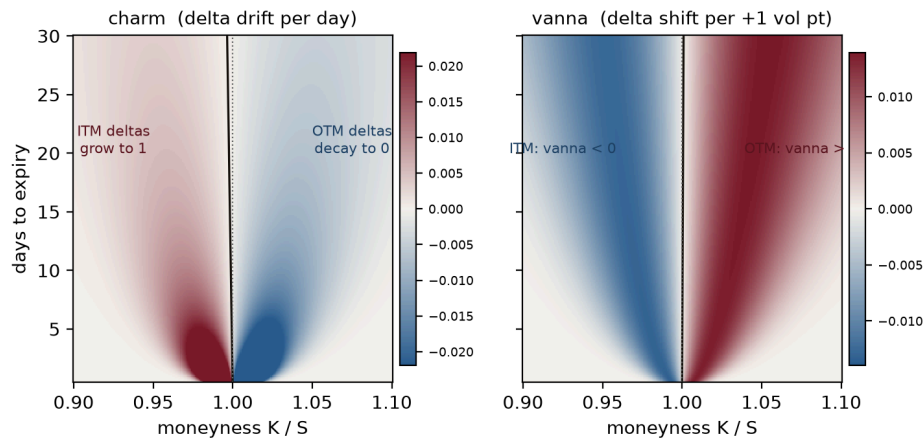
Gamma is only the first of the re-hedging channels. The dealer's delta drifts even when the underlying does *not* move, because the options age. That drift is *charm*, and in the 0DTE era it turned from a textbook curiosity into a driver.

6.1 Charm as the time-decay of delta

Charm is the partial derivative of delta with respect to the passage of time — how much a contract's delta changes per day, holding the underlying and volatility fixed. Its intuition is clean: as expiration approaches, an in-the-money option's delta migrates toward 1 (for a call) and an out-of-the-money option's delta migrates toward 0. Delta is being pushed toward its terminal step-function shape purely by the clock. Because the dealer hedges delta mechanically, any *deterministic* change in delta forces a *deterministic* hedging trade — even on a flat day. (Natenberg 2015)

The sign structure is the useful part. For a call, charm is positive just below the money (its delta is still rising toward 1) and negative just above (its delta is decaying toward 0); the boundary sits near $d_2 = 0$, slightly above the at-the-money strike. Figure 5 (left) maps it. The magnitude is where the story lives: far from expiry charm is negligible — about ± 0.001 of delta per day at 30 days to expiry — but into the final day it explodes, reaching $+0.118$ for a slightly in-the-money call and -0.125 just out-of-the-money at one day to expiry. That is a hundred-fold amplification of the deterministic delta drift as expiry approaches.

BSM charm and vanna for a call: sign flips at the zero contour near $d_2 = 0$, magnitude explodes into expiry



BSM, $\sigma=0.18$, $r=0.04$, $q=0.012$; values clipped at the 2nd/98th percentile; put maps are visually identical (difference $O(q)$).

Figure 5. Black-Scholes charm (left, delta drift per calendar day) and vanna (right, delta shift per +1 vol point) across moneyness and time to expiry ($\sigma=0.18$, $r=0.04$, $q=0.012$; no market data). The black zero contour sits near $d_2 = 0$, just above the money; both greeks change sign across it. Charm is negligible far from expiry — about ± 0.001 per day at 30 days — and explodes into the last day ($+0.118$ for a slightly in-the-money call, -0.125 just out-of-the-money at 1 day) — roughly a hundred-fold amplification. Vanna is largest just off the money and decays more gently with tenor. Reproduce with `figures/fig_charm_vanna_maps.py`.

6.2 Why 0DTE turned charm from curiosity into driver

For decades charm mattered only in the last day or two of monthly expirations, on a modest slice of open interest. Two things changed it. First, the magnitude structure above means charm's force is concentrated in the final hours regardless of era. Second — and this is the regime change — the 0DTE ecosystem now parks an enormous, concentrated block of open interest at strikes that are *always* within a day of expiry. Every session, a large near-the-money book runs through the steep part of the charm surface between the open and the close.

The result is a *deterministic, time-of-day re-hedging flow*: if the dealer's net position and its sign are known, the direction and rough magnitude of the charm-driven hedge into the close are, in principle, predictable from the clock alone, because charm is a function of time, not of price. This is the mechanical basis for the widely observed late-day drift and end-of-day imbalance on expiration-heavy sessions. (JPMorgan/dealer desk commentary)

The timing is the part worth being precise about, because it is what makes charm distinctive. Charm accrues with the calendar, but the *urgency* of hedging it concentrates late in the session: with little time left, the delta drift per remaining hour is steepest, and the hedge that could be spread across a long-dated position must be executed before the same-day contracts settle. The empirical footprint is therefore a clustering of charm-driven hedging into the final hour or two before the cash close and into settlement — historically a monthly-OPEX phenomenon, now a feature of *every* 0DTE session because the near-expiry book is rebuilt daily. The structural novelty is that this flow has a *known clock*. Unlike the gamma flow, whose size and direction depend on a price move that has not happened yet, the charm flow is a smooth,

deterministic function of time; given the standing open-interest distribution and a positioning convention, its cumulative drift over the day is computable *in advance*. That is what turned charm from a footnote into something desks model explicitly. Two erosions temper the exploitability, and both recur throughout this paper: any participant who can compute the same clock will pre-position against the predictable part, arbitraging it toward zero; and the *direction* of the whole thing still hangs on the assumed sign of the dealer position, so a charm forecast inherits the identification problem of Section 3 wholesale. The honest boundary is the same one that dogs every claim in this paper: the *direction* of the charm flow depends on the *sign* of the dealer position (Section 3), which is assumed, not measured; and the tradable residual after other participants pre-position is an empirical question, not a mechanical certainty. The mechanism is real; its exploitability is a hypothesis for Section 10.

6.3 Two charm clocks: the monthly pulse and the daily drumbeat

The 0DTE era did not replace the old charm regime; it added a second one alongside it, and the two coexist with different signatures. The *monthly pulse* is the legacy phenomenon: the third-Friday standard expiration still concentrates the single largest block of open interest of any expiry, so its charm-driven re-hedging into settlement is a large, once-a-month event — on the cached snapshot the next-day September monthly OPEX alone carried 24.3% of all open interest (Section 8), dwarfing any single 0DTE session's standing book. The *daily drumbeat* is the new phenomenon: every session now has an expiry, so the steep part of the charm surface is traversed *every* afternoon, but by a book that is smaller in standing OI and largely built and destroyed within the day. The practical difference is one of predictability and size. The monthly pulse is anchored to a known calendar date and a large, days-old open-interest distribution that is visible in advance, making its charm flow the more forecastable of the two; the daily drumbeat is anchored to an OI distribution that does not exist until the session builds it, so its charm flow can only be estimated with intraday data and inherits the 0DTE identification problem in full. A complete charm model needs both clocks, and needs to keep them separate: averaging a forecastable monthly event with an intraday-built daily one produces a number that describes neither.

7 Vanna flows

The third re-hedging channel is driven not by price and not by time but by *implied volatility*. It is the channel that couples the options complex to the volatility complex, and it is the natural bridge to the volatility-surface paper.

7.1 Vanna: an IV shock becomes a delta shock

Vanna is the sensitivity of delta to implied volatility — how much a contract's delta shifts for a one-point change in IV, holding price and time fixed. It is the same cross-derivative as the sensitivity of vega to the underlying, and its economic content is that *a change in the volatility surface re-prices delta across the whole chain even when the index has not moved*. Because the dealer hedges delta, an IV move alone forces a hedging trade. (Natenberg 2015)

Figure 5 (right) maps the sign structure. For a call, vanna is negative just in-the-money and positive just out-of-the-money, with the zero contour again near $d_2 = 0$; it is largest just off the money and decays more gently with tenor than charm does (about ± 0.013 of delta per vol point at one day to expiry, ± 0.004 at thirty days). Vanna is therefore a *whole-surface* phenomenon, not a last-hours one: it matters wherever there is open interest and wherever IV can move, which is everywhere.

7.2 The vol-spike/rally coupling and the VVIX gate

The market-level story is the *vanna rally / vanna decline* coupling. Under the standard index positioning — dealers short downside puts — a *fall* in implied volatility (a calmer tape) reduces the delta the dealer is short and forces it to *buy* the underlying to stay hedged; a *rise* in IV forces selling. This produces a systematic tailwind to price when volatility is declining and a headwind when it is spiking, mechanically linking the direction of IV to the direction of the hedging flow. (JPMorgan/Kolanovic; SpotGamma-style VEX models)

Two honesty flags are essential here, and both come straight from the project's own evidence discipline. First, the *competing explanation*: IV tends to fall when the market rises anyway — the *leverage effect* — so an observed "IV down, price up" co-movement is *not* by itself evidence of a vanna hedging flow; the leverage effect predicts the same correlation with no dealer in the story. Distinguishing the two requires isolating the *incremental* hedging flow, which needs a clean IV surface the free pipeline does not have. Second, the *evidence tier*: the vanna-rally framework rests on Tier-B practitioner research and Tier-C vendor VEX products, and those vendors *cite each other* — multiple vendor "confirmations" are not independent evidence. The one robust, independent anchor (demand-based option pricing) supports the premise that persistent net demand moves prices and IVs; it does *not* validate the specific VEX-level mechanism. (Garleanu, Pedersen & Poteshman 2009)

A cautionary case keeps the tailwind in proportion. A prominent, widely-circulated late-2018 constructive call rested partly on positioning and flow arguments of this family — a vanna/gamma tailwind into year-end — and was then overwhelmed by the December 2018 drawdown. The lesson is not that the flow was imaginary; it is that a hedging *tailwind* is a second-order push that a first-order shift in fundamentals or risk appetite can swamp in an afternoon. Vanna belongs in the conditioning set, not in the trigger. It sizes and tilts a view that must come from somewhere else; used as a standalone timing signal it will, sooner or later, meet a day when the flow was real and irrelevant.

There is also a *measurement* trap specific to vanna, beyond the identification problem it shares with gamma: **double-counting**. The gamma and vanna re-hedges are not independent flows to be added; they are two projections of the *same* delta drift, one attributed to the price move and one to the accompanying IV move, and price and IV move together. Summing a "GEX signal" and a "vanna signal" therefore risks counting one hedging trade twice. The only honest test of whether vanna adds anything is *incremental*: does the vanna term improve a model that already contains the gamma term and the leverage-effect co-movement? If it does not survive that control, it was never a separate flow — it was gamma wearing a volatility label. This is the design constraint Section 10 imposes on the vanna hypothesis.

The practical synthesis practitioners use — and the honest way to state it — is that vanna dominates when the *volatility of volatility* is elevated: large, fast IV moves are what make the vanna re-hedge big relative to the gamma re-hedge. VVIX (the vol-of-vol) is therefore the natural *gate* for when the vanna channel is worth watching, and the bridge to the volatility-surface material is exactly this: the vanna flow is a volatility-complex phenomenon dressed as an equity flow.

7.3 The three flows are one hedge

It is easy to lose the forest for the trees, so it is worth stating plainly that gamma, charm, and vanna are *not* three separate flows. They are three partial derivatives of the *same* object — the dealer's aggregate delta — with respect to three different variables: the underlying price (gamma), the passage of time (charm), and implied volatility (vanna). On any given day the dealer executes *one* net hedging trade, and to first order that trade is the total differential of its delta,

$$d\Delta \approx \Gamma \cdot dS + \text{charm} \cdot dt + \text{vanna} \cdot d\sigma.$$

Decomposing it into three named flows is analytical convenience, and a useful one, because the three terms have different *clocks* — gamma is price-contingent and fires whenever the market moves, charm is deterministic in time and concentrates into expiry, vanna is IV-contingent and dominates when the vol-of-vol spikes — so knowing which term is largest tells you which variable to watch. But two dangers follow directly from the decomposition, and both have appeared already. The terms are *not independent*: dS , dt , and $d\sigma$ co-move (price up tends to come with IV down and time always passes), so adding three separately-estimated signals double-counts one physical hedge. And the *same assumed sign* of the dealer position multiplies all three terms, so an identification error is not three independent mistakes that might partly cancel — it is one mistake propagated three ways, moving gamma, charm, and vanna estimates together in the wrong direction. The disciplined object of study is therefore the *net* delta hedge and its total differential, not a portfolio of three independently-marketed indicators. When a service sells a GEX line, a charm map, and a vanna model as three confirmations, remember they are three views of one number with one sign in front of it.

8 The 0DTE ecosystem

Every estimator in Section 4 and every mechanism in Sections 5–7 assumes something that 0DTE options quietly break: that the open interest an end-of-day snapshot records is the book that gets hedged. This section shows, on the cached snapshot, how false that has become, and what it does to the interpretability of a daily GEX number.

8.1 Where the activity actually lives

Figure 6 splits the SPX chain by days to expiry into two shares: the share of the *day's volume* and the share of *open interest*. The wedge between them is the whole story. Same-day (0 DTE) contracts are 58.6% of the day's 4.8 million traded contracts but only 1.4% of the 25.9 million open interest. The next-day expiry — here the September monthly OPEX, one day out — carries 24.3% of OI. Volume concentrates violently at zero DTE while open interest lives far out on the curve (53% of it beyond 63

days). On this snapshot the 0–1 DTE bucket alone is 75.7% of the day's volume. This is consistent with CBOE's published trend, which puts 0DTE at roughly 62% of SPX option volume year-to-date through August 2026, up from about 17% in 2020. (CBOE 2023)

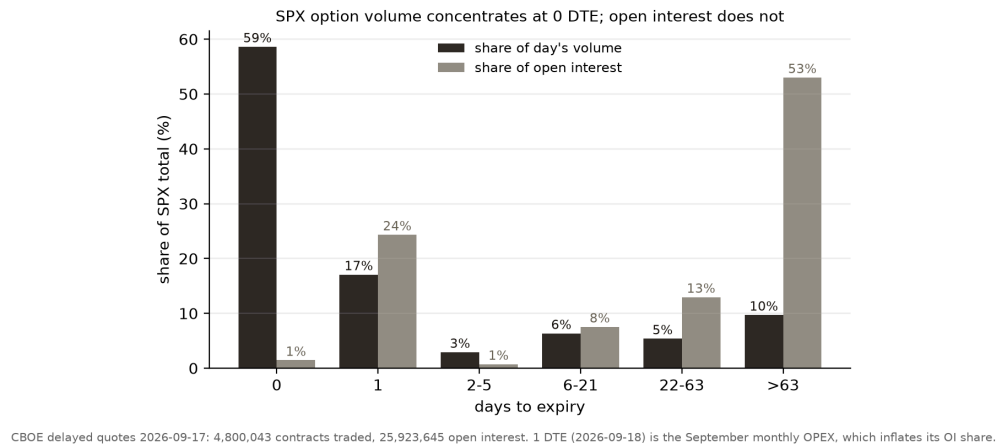


Figure 6. Where SPX option activity lives, from the cached CBOE snapshot (2026-09-17): share of the day's volume versus share of open interest, by days to expiry. Same-day (0 DTE) contracts are 58.6% of the day's 4.8 million traded contracts but only 1.4% of the 25.9 million open interest; the next-day September monthly OPEX (1 DTE) carries 24.3% of OI. Volume concentrates at zero DTE while open interest lives far out (53% beyond 63 days) — the wedge is the intraday measurement gap: positions opened and closed the same day never enter any end-of-day open-interest snapshot. Reproduce with `figures/fig_0dte_share.py`.

8.2 Intraday greeks and a dealer inventory measured in hours

The reason 0DTE matters for hedging is that its greeks scale inversely with the square root of time to expiry. Gamma near the money grows like $1/\sqrt{T}$, so a 0DTE at-the-money option has enormous gamma per contract; charm explodes on the same day (Section 6); theta decays the whole premium within the session. A 0DTE book is therefore a large, fast-moving gamma and charm exposure that *is created and destroyed within a single trading day*. The dealer's inventory half-life for this part of the book is measured in *hours*, not days: the position that must be hedged at 10:00 may not exist at 15:00, and the position hedged into the close expires at the close.

This intraday life cycle has a consequence that quietly poisons every historical GEX pipeline. A 0DTE contract's open interest is not a *standing* quantity that a daily snapshot samples; it is *built during the session* — opened in the morning, traded through the day, and largely extinguished at the close. An end-of-day OI snapshot therefore records the *wreckage* of the 0DTE book, not the live exposure that actually drove hedging while the session was open. The only faithful measurement of 0DTE gamma is *intraday and timestamped*: one must know the OI and the sign of the flow *at the hour of interest*, because both change within the day. Any 0DTE hypothesis that leans on end-of-day data is measuring the wrong object, and any pre-2022 study — built in a world where the near-expiry book was a monthly, not a daily, phenomenon — must earn its transfer across the structural break rather than assume it.

The concentration also sharpens *pin risk* into the final minutes. As a 0DTE at-the-money option approaches settlement, its gamma per contract diverges and its delta becomes a near-binary function of

whether the index finishes a point above or below the strike; the hedge required to stay neutral through that transition is enormous relative to the premium at stake, and it must be executed in a shrinking window. Whether the resulting flow pins the index or whips it depends, once again, on the sign of the aggregate dealer gamma — long gamma pulling toward the strike, short gamma pushing away — which is exactly the sign the end-of-day data cannot supply.

8.3 What 0DTE does to every Section 4 estimator

Return to the measurement gap. An end-of-day OI snapshot — the only free historical input that could ever build a GEX series — records 0DTE contracts at 1.4% of OI while they were 58.6% of the day's flow. The gamma that actually drove hedging during the session is precisely the gamma that has *expired and left no trace in OI* by the time the snapshot is taken. Three consequences follow, and each attacks an earlier estimator directly:

- **The OI-weighted estimator (E1) systematically misses the dominant intraday book.** A GEX built from end-of-day OI is a GEX of the *longest-dated* half of the market — the half that hedges slowly — while the fast, near-expiry gamma that moved the tape is invisible.
- **The volume-weighted estimator (E2) captures the 0DTE flow but cannot sign it.** Volume records that the 0DTE contracts traded, not whether customers were net buyers (dealer short gamma) or net sellers (dealer long gamma) — and in 0DTE both are large. The identification problem of Section 3 is at its worst here.
- **The daily GEX number loses a stable meaning.** If most of the day's hedging pressure comes from a book that is born and dies intraday, a single end-of-day gamma figure is not a summary of the day; it is a snapshot of the slow book after the fast book has gone home. Any 0DTE-era claim requires *intraday timestamped* OI, which the free pipeline does not provide, and pre-2022 gamma studies do not automatically transfer across this structural break.

The upshot is not that 0DTE makes GEX wrong, but that it makes the *daily, end-of-day* GEX far less interpretable than its clean line suggests, and it does so exactly in the market that watches GEX most.

8.4 The interpretability verdict for a daily GEX number

It helps to end this section with a plain verdict, because the diagnosis is easy to mistake for nihilism, which it is not. A daily, end-of-day GEX number in the 2025-era index market is best understood as a *lagging measure of the slow book* — the longer-dated open interest that hedges gradually — with the fast, near-expiry gamma that actually moved the tape mostly invisible to it. That is not nothing: the slow book is real, its flip and walls are genuine features of a real gamma profile, and they carry information about where anti-cyclical hedging would resist a move *if* the standard sign is right. But it is far less than the clean daily line advertises, and it should be used accordingly: as a *regime prior* carried with wide error bars and an explicit sign caveat, re-derived intraday from timestamped data whenever the question is about the current session, and never as a precise, static level that the day must respect. The honest one-line summary is that end-of-day GEX describes the book *after the fast money has gone home*, and any use of it has to price that in.

9 What GEX cannot do

The preceding sections built the mechanism carefully and tested what free data allow. This one is the counterweight: an explicit catalogue of the failure modes, so the paper cannot be quoted as an endorsement of the thing it is dissecting.

9.1 Identification drift

The sign convention of Section 3 was calibrated to a flow regime — institutional overwriting and protective-put buying — that is a historical contingency, not a constant. As the customer mix drifts (retail call buying in 2021, two-sided 0DTE premium selling since 2022), the assumed dealer sign drifts away from the true one, silently. A GEX product that fixes the convention and never revisits it will keep printing a confident regime call while the assumption underneath it decays. There is no free-data alarm for this drift, which is why the number should be read as *a scenario conditional on an assumption*, not a measurement. (Section 4.3 shows the sign is not robust even on one day)

9.2 Flow versus stock

GEX is a *stock* — a snapshot of standing exposure — but the thing that moves prices is a *flow*, the actual hedging trades executed over an interval. The two are related but not identical: a large gamma stock produces no flow on a flat, low-volatility day, and a modest stock can produce a large flow in a fast tape. Confusing the standing exposure with the executed flow is the error behind statements like "high gamma at this strike means price will go there" — high gamma means large *potential* hedging *if* price moves, not a force that materializes on its own. A residual idea spun off from this paper's own early work makes the distinction concrete: high at-the-money gamma near expiration reliably implies high *hedging volume*, which is a flow claim about turnover, and says nothing directly about the *direction* or level of volatility. Flow claims and stock claims need separate tests.

9.3 Endogeneity: does GEX cause damping, or proxy for it?

The hardest problem is causal. The proxy study of Section 5.3 makes it vivid: low-VIX days are calm, and low-VIX days are also (under the standard convention) positive-gamma days, so "positive gamma predicts low realized volatility" may be nothing more than "calm days are calm." The gamma regime and the volatility level are *jointly determined* — low realized volatility invites more call overwriting, which raises GEX, which is then credited with the calm it merely accompanied. Reverse causality (vol → positioning) and common causes (a quiet macro regime driving both) are not nuisances to footnote; they are candidate *full explanations* of the correlation. Breaking them requires conditioning on the gamma sign while *controlling* for lagged realized volatility and the VIX level — which is precisely the regression Section 10 pre-registers, and precisely what the free proxy cannot do.

9.4 Competing explanations for the same stylized facts

The two facts GEX is usually invoked to explain — that volatility clusters and that down moves are more volatile than up moves — both have older, simpler explanations that make no reference to dealers at all:

- **Volatility clustering** is a universal feature of asset returns, captured by GARCH-type models decades before GEX existed. The monotone magnitude channel of Figure 3 is fully consistent with clustering alone; it is *not* evidence for a gamma mechanism.
- **The leverage effect** — volatility rising as prices fall — explains the negative-gamma "amplification below the flip" pattern without any hedging flow, because falling prices raise realized and implied volatility for balance-sheet reasons that predate options market makers.

To attribute a stylized fact to dealer hedging rather than to clustering or leverage, one must show *incremental* explanatory power after those alternatives are included as controls, and ideally exploit a variable that shifts dealer gamma *without* shifting the alternatives — a genuine identification strategy, not a suggestive correlation. The literature that does this carefully (the illiquidity-conditional intraday results of Barbon & Buraschi) finds a real but conditional effect; the free-data proxy here cannot, and does not claim to.

9.5 The direction-check failure, concretely

The most expensive error in this paper's own research history was not a bad model; it was a sign. An early formulation of the core hypothesis stated that large gamma near expiration *increases* volatility, justified by "anti-cyclical hedging." Confirming evidence then arrived — of the right *magnitude* — and was recorded as validation. The evidence in fact showed volatility being *damped*: anti-cyclical hedging (sell strength, buy weakness) dampens by definition, so the original claim was self-contradictory, and the very number cited as support (a lower realized volatility in the high-gamma bucket) pointed the opposite way to the stated direction. Right size, wrong sign, recorded as a confirmation. The fix is a reflex, not a technique: for every piece of evidence, state in one line whether it *supports*, *contradicts*, *duplicates*, or merely *extends* the claim, and when the sign disagrees, flag both the new evidence and the old claim rather than absorb it. GEX, whose entire content is a sign, is exactly the domain where this reflex earns its keep.

9.6 Estimator choice as a hidden researcher degree of freedom

Section 4.3 showed three defensible estimators placing the flip on three sides of spot. That sensitivity is not only a measurement caveat; it is a *data-snooping hazard*, and it deserves to be named as one. A researcher who tests "does the flip predict the next day?" and is free to choose, after seeing the outcome, whether to use OI-weighting, volume-weighting, delta-adjustment, a total-versus-front-month window, or a net-versus-gross sum, is walking a garden of forking paths of exactly the kind the quantitative-methods papers warn against: each construction choice is a parameter fitted, unconsciously, to the sample. Because all of these choices are *defensible*, the temptation to keep the one that "works" is strong and easy to rationalize. The discipline is the same one imposed on any pre-registered test — fix the estimator, the sign convention, the expiry window, and the net/gross choice *before* the out-of-sample data are opened, and report the sensitivity of the result to those choices as part of the finding, not as a menu from which the best answer is served. A GEX result that survives only under one hand-picked construction is an order statistic, not a discovery.

9.7 The signal decays as it is watched

GEX is unusual among flow measures in being *freely and widely published*. That creates two reflexive problems for anyone hoping to trade it. First, when a level like the flip or a wall is watched by many, participants trade *around it directly*, so an apparent confirmation — price respecting the wall — may be a self-fulfilling coordination effect among chart-watchers rather than the dealer hedging mechanism the level is supposed to proxy; the two are observationally similar and require different tests to separate. Second, a genuinely predictable flow is a public edge, and public edges are arbitrated: the pre-positioning that Sections 6 and 7 flagged for charm and vanna generalizes to any documented GEX relationship, so even a real historical effect need not persist once it is known. The consequence for practice is that a GEX-based claim cannot be validated once and shelved; it must be re-tested on rolling out-of-sample data, with the null that the effect has decayed taken seriously. A mechanism can be real, its historical footprint genuine, and its forward edge already gone — three statements this paper holds simultaneously without contradiction.

10 From mechanics to testable hypotheses

The purpose of the preceding nine sections was to earn the right to state a small number of claims precisely enough to be wrong. This section does that — translating each mechanism into a pre-registered, falsifiable test — and then describes the fully reproducible free-data pipeline end to end, so that the study which this paper *cannot* run on free snapshots can be run by anyone with an end-of-day options dataset.

10.1 Four claims, four pre-registered tests

Each mechanism becomes a confirmatory hypothesis with a sign, a control set, and an out-of-sample split fixed before the data speak. The hand-off is to the quantitative-methods papers (pre-registration in the backtesting paper, multiple-testing correction in the overfitting-control paper).

- **Regime / realized volatility (the H001–H002 pair).** *H1*: higher positive dealer GEX on day t predicts *lower* absolute return on day $t+1$, and negative GEX predicts *higher*, after controlling for lagged realized volatility, the VIX level, and an OPEX dummy. *H0*: GEX carries no incremental information about next-day realized volatility once those controls are included. Primary test: the sign and significance of the GEX coefficient in that regression, plus a quartile-monotonicity check on the GEX-sorted next-day realized volatility. Significance via block-bootstrap and Newey-West standard errors; **direction check** built into the decision rule (a significant coefficient of the wrong sign *refutes*, it does not support); multiple-testing correction (Benjamini-Hochberg) across the planned family. In-sample design and threshold selection on the earlier period, out-of-sample opened once at the end. *This is the test that has not been run here, because it needs the historical OI the free pipeline lacks; nothing in this paper confirms it.*
- **Charm drift (H003).** *H1*: on high-0DTE-OI sessions, the sign of the late-day return imbalance matches the sign of the modelled charm hedging flow implied by the standing OI and an explicit positioning convention. *H0*: no relationship after controlling for market-on-close imbalance.

Requires *intraday* OI and timestamps; falsifiable, and conditional on the dealer gamma sign (long → pin, short → anti-pin).

- **Vanna coupling (H004).** *H1*: the *incremental* underlying hedging flow attributable to an IV move — after removing the leverage-effect co-movement — has the sign the standard positioning predicts, and is larger when VVIX is elevated. *H0*: the IV-price co-movement is fully explained by the leverage effect with no incremental hedging flow. Requires a clean IV surface; the double-counting against gamma must be handled by testing *incrementality*, not by adding a separate vanna signal.
- **Pinning (already tested here, in the null).** Section 5.4 pre-committed one rule and found no index-close pinning; the single-stock, deliverable version remains the literature's positive result. This is the template: one rule, fixed in advance, reported whatever it shows.

To make the first and most important of these concrete, here is the frozen pre-registration for the regime test — the one this pipeline cannot yet run, because it needs the historical open interest the free feeds lack. It is written in the same append-only form the quantitative-methods papers require, so that when an end-of-day OI feed is added the test is executed exactly as specified, not re-negotiated to fit the answer:

```

# Pre-registration – GEX regime vs next-day realized volatility

## Confirmatory hypothesis (exactly one)
- H1 (directional): higher positive dealer GEX on day t predicts LOWER absolute
  return on day t+1, and negative GEX predicts HIGHER; the GEX coefficient
  is negative and significant AFTER controls.
- H0: GEX carries no incremental information about  $|r(t+1)|$  once lagged realized
  volatility, the VIX level, and an OPEX dummy are included.

## Data & sample (fixed)
- GEX series: rebuilt daily from end-of-day strike-level OI (e.g. optionsDX EOD),
  standard sign convention, per-1% dollar-gamma, total across listed expiries.
- Prices / volatility: SPX and VIX daily closes (free feeds).
- Period: <start> - <end>, fixed before any run.

## In-sample / out-of-sample split (fixed dates, not fractions)
- In-sample (design & any threshold selection): dates <= <split>.
- Out-of-sample (opened once, at the end): dates >= <split>.

## Model & metric
- Primary: sign and significance of  $b_{\text{GEX}}$  in
 $|r(t+1)| \sim b_{\text{GEX}} \cdot \text{GEX}(t) + b_{\text{rv}} \cdot |r(t)| + b_{\text{vix}} \cdot \text{VIX}(t) + b_{\text{opec}} \cdot \text{OPEX}(t) + e.$ 
- Secondary: monotonicity of  $|r(t+1)|$  across GEX quartiles; Spearman( $\text{GEX}(t), |r(t+1)|$ ).

## Decision rule (chosen before the data are opened)
- Support H1 iff  $b_{\text{GEX}} < 0$  with block-bootstrap / Newey-West  $p < 0.05$  IN-SAMPLE,
  the sign holds OUT-OF-SAMPLE, AND the quartile pattern is monotone.
- Direction check: a significant  $b_{\text{GEX}} > 0$  REFUTES the claim, it does not support it.

## Multiple-testing correction
- Family = {regression, quartile test, Spearman, sign-flip event study}: Benjamini-Hochberg.
- Estimator variants (Section 4.3) fixed in advance; their sensitivity is REPORTED, not selected on.

## Frozen on
- Date: <YYYY-MM-DD> · Commit: <hash> · Seed: <n>

```

The single most important field is the control set. Regressing next-day absolute return on GEX *alone* would recover the SqueezeMetrics quartile result and mean nothing, because — per Section 9.3 — it would be conditioning on the volatility level under another name. Only the *incremental* coefficient, after lagged realized volatility and the VIX level are already in the model, can distinguish a dealer-hedging effect from volatility clustering and the leverage effect. This is the test whose in-house run is still pending in this pipeline; nothing in this paper reports its result, and the honest status of the whole regime story is "well-motivated, literature-backed, and not yet confirmed on own out-of-sample data."

10.2 The reproducible free-data pipeline, end to end

The construction side of this paper is fully reproducible today; the historical study is reproducible by anyone who adds an end-of-day OI feed. The pipeline, stage by stage:

1. **Chain download.** Pull the SPX option chain from the CBOE public delayed-quotes endpoint (cdn.cboe.com/api/global/delayed_quotes/options/_SPX.json, User-Agent header required) and

cache the raw JSON. This gives per-contract strike, expiry, open interest, volume, implied volatility, and greeks.

2. **GEX build.** For each live contract (same-day settled expiry excluded), compute dollar gamma per 1% move as $\Gamma \cdot \text{OI} \cdot M \cdot S^2 \cdot 0.01$, apply an *explicit, stated* sign convention, and sum. Recompute gamma from each contract's IV under Black-Scholes-Merton as a cross-check (here it matched the vendor gamma to 0.1% near the money). Locate the flip by re-evaluating the sum across a spot grid, and the walls as the extreme per-strike strikes. Report the answer under *at least* the standard, volume-weighted, and delta-adjusted estimators, so the sign sensitivity of Section 4.3 is visible rather than hidden.
3. **History (the part free snapshots cannot supply).** Replace the single live snapshot with a per-day end-of-day OI history — for example optionsDX EOD files — rebuild the daily GEX series under the fixed convention, and merge with SPX and VIX daily data (Yahoo/FRED).
4. **Regime study.** Run the pre-registered regression and quartile test of Section 10.1 on that series, with the in-sample/out-of-sample split, block-bootstrap inference, direction check, and multiple-testing correction fixed in advance. Log the result append-only, whatever it shows.

One point of reproducibility discipline closes the pipeline. Because a GEX edge decays as it is watched (Section 9.7), the study is not a one-time verdict but a standing process: the result log is append-only, the out-of-sample window rolls forward, and the effect is re-tested on genuinely new data with the null that it has decayed taken seriously each time. Every stage is pinned — the download endpoint, the sign convention, the estimator, the expiry window, the split dates, the seed, and the commit hash — so that the number a reader regenerates next year is the number this pipeline produced, and any change in it is a change in the world rather than in the code. That reproducibility, not any single figure, is what a free-data treatment of GEX can honestly offer.

The honesty of the whole enterprise reduces to one discipline visible at every stage: state the assumption that supplies the missing sign, show the answer's sensitivity to it, and never let a construction on a live snapshot masquerade as a validated historical relationship. The mechanism is real, the literature is genuine, and the confirmatory test on one's own free-data pipeline is designed, pre-registered, and — pending the OI history — still open.

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis
Three GEX estimators place the gamma flip at 7,656 / 7,633 / 7,641 and disagree on the sign of net dealer gamma at spot (−28 / +8 / −11 dollar-billion per 1%)	Own reproducible computation — figures/fig_gex_estimators.py
Strike-level GEX profile: call wall 7,800, put wall 7,500, flip ~7,656 (+0.24%); GEX at spot straddles zero — +\$5.4bn per 1% from CBOE-supplied gamma vs −\$28bn from recomputed BS gamma (E1), the snapshot sitting on the flip	Own reproducible computation — figures/fig_gex_profile.py
Charm and vanna sign maps flip near $d_2=0$ and explode into expiry (charm ± 0.001 at 30 days to +0.118/−0.125 at 1 day)	Own reproducible computation — figures/fig_charm_vanna_maps.py
VIX-tercile proxy: mean absolute next-day return rises monotonically in and out of sample; the high-VIX autocorrelation (−0.22) collapses out of sample (−0.02)	Own reproducible computation — figures/fig_regime_proxy.py; OOS split
A pre-committed OPEX pinning test on cash-settled SPX returns a null (6.28 versus 6.39 points, $z=-0.37$, $p=0.36$)	Own reproducible computation — figures/fig_pinning_test.py
0DTE contracts are 58.6% of the day's volume but 1.4% of open interest; 53% of OI sits beyond 63 days	Own reproducible computation — figures/fig_0dte_share.py
Dealers hedge delta mechanically and warehouse gamma, vega, and theta	Literature — (Natenberg 2015; Garleanu, Pedersen & Poteshman 2009)
Negative dealer gamma is associated with intraday momentum and positive with reversal, conditional on illiquidity	Literature — (Barbon & Buraschi 2021)
Persistent net customer demand moves option prices and implied volatilities	Literature — (Garleanu, Pedersen & Poteshman 2009)
Pinning at high-open-interest strikes is a single-stock and futures result, conditional on the dealer-gamma sign	Literature — (Ni, Pearson & Poteshman 2005; Golez & Jackwerth 2012)
The standard sign convention (dealers long calls, short puts)	Practitioner consensus — (SqueezeMetrics 2017)
The vanna-rally coupling and the late-day charm re-hedging drift	Practitioner consensus — not independently verified

References

- Barbon, A., & Buraschi, A. (2021). *Gamma Fragility*. University of St. Gallen, School of Finance Research Paper No. 2020/05. Open access: <<https://ssrn.com/abstract=3725454>>. — The strongest peer-review-track anchor for the regime mechanism: in an equity-options panel, negative ex-ante dealer gamma imbalance is associated with intraday momentum and positive imbalance with reversal, conditional on underlying illiquidity — the sign pattern and the conditionality the mechanism predicts.

- Ni, S. X., Pearson, N. D., & Poteshman, A. M. (2005). Stock price clustering on option expiration dates. *Journal of Financial Economics*, 78(1), 49–87. — The canonical pinning result: single-stock prices cluster near high-open-interest strikes on expiration Fridays, via a deliverable-settlement mechanism that does not automatically transfer to a cash-settled index.
- Ni, S. X., Pearson, N. D., Poteshman, A. M., & White, J. (2021). Does option trading have a pervasive impact on underlying stock prices? *Review of Financial Studies*, 34(4), 1952–1986. — Estimates that option-market delta-hedging accounts for a non-trivial share (on the order of a tenth) of daily absolute stock returns, giving the hedging-flow channel independent, peer-reviewed magnitude.
- Garleanu, N., Pedersen, L. H., & Poteshman, A. M. (2009). Demand-based option pricing. *Review of Financial Studies*, 22(10), 4259–4299. — Establishes that persistent net customer demand moves option prices and implied volatilities in the direction dealer inventory implies; supports the *positioning premise* behind GEX, not the specific GEX-to-realized-volatility mechanism.
- Golez, B., & Jackwerth, J. C. (2012). Pinning in the S&P 500 futures. *Journal of Financial Economics*, 106(3), 566–585. — Finds pinning in the S&P 500 *futures* that is conditional on the sign of dealer gamma (long gamma pins, short gamma anti-pins), sharpening why the cash-index test can come back null.
- Baltussen, G., Da, Z., Lammers, S., & Martens, M. (2021). Hedging demand and market intraday momentum. *Journal of Financial Economics*, 142(1), 377–403. — Links dealer hedging demand to a systematic intraday-momentum pattern in index markets, complementing the single-stock evidence at the index level.
- SqueezeMetrics (2017). *Gamma Exposure (GEX)*. White paper. Public PDF: <https://squeezemetrics.com/monitor/download/pdf/white_paper.pdf>. — The origin of the standard sign convention (dealers long calls, short puts) and of the popular GEX line; cited here critically, as the source of the positioning assumption whose fragility Sections 3–4 are built to expose, not as validated evidence.
- Cboe Global Markets (2023). *Evaluating the market impact of SPX 0DTE options*. Cboe Insights. <<https://www.cboe.com/insights/posts/volatility-insights-evaluating-the-market-impact-of-spx-0-dte-options/>>. — Exchange-published data and analysis on the growth of 0DTE to a majority of SPX option volume, the basis for the structural-break caveat in Section 8.
- Natenberg, S. (2015). *Option Volatility and Pricing* (2nd ed.). New York: McGraw-Hill. — Standard practitioner-textbook treatment of dynamic delta-hedging and the higher-order greeks (gamma, charm, vanna) used throughout Sections 2, 6, and 7.

All references above are publicly accessible: one is an open-access SSRN working paper (direct link given), one is a public exchange research post and one a public white paper (both linked), five are peer-reviewed journal articles, and one is a published textbook. No proprietary, licensed-vendor, course, or trading-academy material is cited; where the popular presentation of GEX depends on such a product, the quantity is rebuilt from free data instead.